# PROPOSAL

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# Proposal:

Rolsa Technologies are a local green technology company aiming to create a digital solution providing information about green energy products on the market and how to reduce their carbon footprint as well as allowing customers to schedule consultations and installations and calculate their carbon footprint.

As a company offering green technology products and services, Rolsa's solution should address the needs of users concerned with being environmentally conscious. It must incorporate proven strategies of similar businesses in the field, while introducing innovative features which differentiate them from the competition. This means that it must be stylised with the company goals and

objectives in mind and contain features consistent with the requirements laid out in the set brief, as well as other relevant inclusions. Information must be made accessible to all users in a clear manner, and the solution must contain pages detailing the various services available to be booked or purchased. Finally, to satisfy the need of carbon footprint calculation, the solution should implement realistic and accurate formulae and allow necessary data to be input by users to calculate their footprint.

In order to meet these needs, my solution will be a database driven online website coded in PHP, HTML and SQL with various API integration included. My site, to be accessible to a wide range of potential users, will be functional on both mobile and desktop devices with WCAG compliant accessibility features available. My website will include objectives included in the brief such as various information pages on the company, what they offer and what new green products exist on the market in a clear and easily accessible manner, a carbon footprint calculator complete with personalised advice on how to reduce said footprint and a scheduling page for consultations and installations.
As well as this, my proposed solution will also include the additional features from the client's market research: account registration and an energy usage tracking tool. As previously hinted at, my account registration system will be implemented using PHP database. It is therefore important that validation and sanitisation techniques are implemented holistically into my solution to ensure a safe website to maintain and build upon the company's reputation and avoid cyberattacks.
My website will also include additional features which I have deemed necessary and beneficial for the solution. These will include an affiliate program and a chart showing the amount solar panels installed over time. This chart will be used on the home page to demonstrate the company's commitment to green technologies and environmental sustainability.

Throughout the following document, I will detail each step of my solution and explain thoroughly the purpose of each design choice I have made.

## User Stories:

User stories serve as an informal short representation of the needs of a diverse set of users to effectively communicate the needs which the website must meet. Here, they will be used to demonstrate the purpose of many requirements, being both functional and non-functional.

| As a [user] | I want to | So that | Acceptance criteria |
|---|---|---|---|
| **Business representative** | Access integrated services and apply to affiliate programs as well as contact the business via email or phone | I can form business partnerships and use specialised business services | The website must include a contact page. This should be easy to navigate to. The website should also offer affiliate programs to other businesses. |
| **Individual user** | Access information and | I can be well informed of the | The website must allow for booking |

| | | | |
|---|---|---|---|
| | have the ability to schedule consultations and receive quotes on the price I will need to pay for the associated services | services offered by the business and know how much I will need to pay and discuss the reasons of this pricing and any other implications of the respective service | of consultations and therefore have an account registration system to manage said consultations. The website could also have a quote estimation feature, for example calculating cost of solar panels based on roof area. |
| **Public sector agency worker** | Access a compliance statement, information on the site and evidence of data protection | I can verify the legality of the company and the purpose of their operations | The website must include a compliance page, a cookie policy page, various information pages, a ToS page and a privacy policy. |
| **Shareholder** | Access strategic updates and information as well as contact information | I can contact the business should I need to and see what new developments are being made | The website should include a newsletter site with new information and company developments. The website must include contact information. |
| **Job seeker** | Access contact information | I can check if there are any availabilities | The website must have a contact information page and could also include a linkedin account link |
| **Researcher** | Access information, metrics and contact details | I can use these facts in my research and ask the company personally about any additional queries | The website must include contact information and a suitable number of metrics, facts and figures as well as qualitive information to help users specifically interested in |

| | | | |
|---|---|---|---|
| | | | research. Facts and figures must be accurate (they would be accurate in a release version of the website). |
| **Accessible user** | Have appropriate accessibility options | I can facilitate my use of the website and relieve complications | The website must follow WCAG 2.2 (or most recent) guidelines to facilitate as many users as possible. |

## Functional requirements:

These are requirements which define what the system must do. This includes specific behaviours, features and functions.

### Home page:

The site must include a home page which has effective navigation to associated pages. I will choose a header for this website's navigation due to other successful businesses in the field adopting a header navigation structure such as Fervo energy and Invenergy. On the mobile version of the website, the header will have a large dropdown menu where users can navigate to other pages.
The home page must also display effective and appealing content to aid user experience. As mentioned previously, I will implement a chart of some kind displaying solar panel installations over time. This will help demonstrate commitment to the company's green ethos and shows reliability. As well as this, the home page will be scrollable and display other meaningful information such as green energy products currently on the market. Most standard pages will share the header for navigation.

### Account registration:

Account registration will consist of a login, signup and accounts page. The login page will take inputs for an email and password with a forgotten password option available. This is so that users can reset passwords and will not be locked out. Users will be able to access most sites logged out as it will not be necessary for users to be logged in and not forcing them to will improve their experience by making it less tedious.

Login process input diagram:

| **Email** | **Password** |
|---|---|

The reset password page will have email, new password and confirm password inputs, ideally this page will be accessed through an emailed link. This will also be

accessible via a link on the accounts page. This makes changing your password voluntarily simpler, improving user experience.

Forgot password / password reset input diagram:

| Email | New Password | Confirm Password |
|---|---|---|

The signup page will take a name, an email a password and a confirmation password. The name has been chosen as an input for personalisation. The signup logic will not allow passwords of an incorrect format (e.g. under 8 characters), it will not allow emails of the wrong format and will not allow an email of an account already in use. All string inputs throughout the program will have proper sanitisation to reduce the cybersecurity threats and prevent vulnerabilities such as SQL injections.

Signup input diagram:

| Name | Email | Password | Confirm Password |
|---|---|---|---|

## Consultation management and account management page:

I have chosen to have these pages as one for simplicity, as they are the only pages which require the user to login to access. Therefore, in the user interface, this page will be linked in place of login options where a user is already logged in. From this page, users will be able to change their name and password. Changing the name will be a simple one input process, since a user can change their name as many times as they wish.

On this page, users can also view their past and future consultations, which will be held either in person or on an external application such as Teams. Before a consultation, a user will have to pay on this site for said consultation. After a consultation has concluded, a user may be given a quote which can be paid on this site. Payments will also be handled externally through apps like PayPal, which will be integrated into the site. This is because there are many legislative complications with storing user bank details. On this page, users can also cancel consultations they have booked and no longer wish to do, however refunds will not be offered.

Name change input diagram:

| New Name |
|---|

## Consultation booking page:

The consultation page will allow users to schedule a consultation. This will be done through a form where the user states a subject and the details as well as their address. The address will use an API to autofill details for the user. The user can then select from a calendar when they are available, with the calendar

displaying the professional's availability. User can be given an estimated quote for the service they request.

### Carbon footprint calculator page:

The carbon footprint calculator page will allow users to get an estimate of their carbon footprint through use of an accurate formula, giving personalised advice on reducing this carbon footprint.

### Accessibility page:

The accessibility page will store a user's preferences with regards to accessibility options as well as documenting the sites following of the WCAG 2.2 legislation. This information will also be available on the compliance page.

### Compliance page:

This page will document how the site follows all relevant legislation, displaying some evidence and description of the process such that any interested parties may access this information and verify the legality of the site. This page will be in the footer of the website.

### Terms of service:

This page will define the websites terms of service to any users who have an interest in seeing them.

### Privacy policy and cookie policy page:

This page will broadly explain how the website collects, uses and protects all personal data as well as the purpose and usage of all cookies on the website (this will be for accessibility and login purposes only). The user may also be able to off cookies for accessibility purposes on this site although the site will say that this is not recommended.

### Contact page:

This page will allow users to see relevant contact information.

## Non-functional requirements:

Non-functional requirements define how a system performs its functions. This means speed, security, maintainability and reliability.

### Security:

The website must be secure. This will mean that it uses input validation, data sanitisation, access control and regular backups. UK GDPR and Data Protection Act mandates appropriate security measures, transparency and data breach management.

### Maintainability:

The solution must be easily maintained by future developers. This requires modular and clean coding standards including organised CSS and HTML, consistency in styling, documentation including regular commenting and routine backups.

### Performance:

Performance attributes focus on speed, scalability and responsiveness. This therefore means that they are measurable constraints. For my performance requirements, the site must handle 20 concurrent users, with pages taking no longer than 2 seconds to load. Performance may be subject to other tests later in development.

### Compliance:

The website must comply with all existing relevant legislation to properly maintain legality.

### Accessibility:

The website must be useable by as many people as possible, therefore it should follow WCAG 2.2 (or the most recent version) as well as taking any further initiative to make the website useable. This means that the website must be perceivable, operable, understandable and robust regarding many types of content. These include text, images, audio, video and many more. The full guidelines can be found at https://www.w3.org/TR/WCAG22/

## Key performance indicators (KPIs):

Key performance indicators are measurable metrics which track how well a site achieves its goals. Some of the following will be utilised and considered for my solution's success:

- Page views
- Conversion rate
- Page load times

While page load times is also a non-functional requirement, it can also be considered a KPI.

## User acceptance criteria:

A user acceptance criterion is a pass-fail test which determines how well the website performs with respect to its functional and non-functional requirements. Here are some criteria the site should meet:

Home page:
- User can navigate to other pages
- User is displayed meaningful content in an understandable way
- User can access information about products currently on the market

Account registration:
- User can navigate back to home page
- User can create an account
- User can login
- User can change their password

Consultation management and account management page:
- User can navigate back to home page
- User can change their name
- User can pay quotes put forward after consultations
- User can pay for future consultations
- User can cancel consultations

Consultation booking page:
- User can navigate back to home page
- User can fill in a form with the subject and details of their consultation
- User can select when they are available from a calendar which shows availability to them
- User can get an estimated quote before booking a consultation

Carbon footprint calculator page:
- User can navigate back to the home page
- User can calculate estimate of carbon footprint
- User can receive advice on reducing carbon footprint

Accessibility page:
- User can navigate back to home page
- User can see how the site follows WCAG guidelines
- User can select accessibility options

Compliance page:
- User can navigate back to home page
- User can see how the website complies with relevant legislation

Terms of service page:
- User can navigate back to home page
- User can read the terms of service of the website

Privacy policy and cookie policy page:
- User can navigate back to home page
- User can read privacy policy

- User can read cookie policy
- User can turn off accessibility cookies

Contact page:
- User can navigate back to home page
- User can access relevant contact information

All pages:
- Consistent navigation and styling
- Follow relevant guidelines
- Load in 2 seconds or less
- Open and function as intended

## Justifications:

### Client and user needs:

My recommended solution will meet the needs of the clients and the users. These needs are as follows:

Provide customers with information about:
- Green energy products currently on the market
- How to reduce carbon footprint

Allow customers to:
- Schedule consultations and installations
- Calculate their carbon footprint

My home page will provide content about energy products currently available on the market to users. This meets the needs of the client in a good way, by having the information easily available on the first site the user sees.
Information on reducing carbon footprints and calculating said footprints have been put into one place. This is a good solution because it allows for more personalised advice and information in reducing carbon footprints.
My solution will contain a site where users can book consultations. These are then paid off on the accounts page to be confirmed. Afterwards, users will be given a final quote for their required instillation, which is also to be paid from the accounts page. I believe that this meets the needs of the client to a good degree and that adding in quotes is a necessary feature for the website as it allows the company to formally charge users for their services.

### Risk mitigation:

The main risks of this solution are cyberthreats to personal data and bank details. As such I will take necessary measures to prevent these risks. Password hashing and salting will be implemented to ensure program security. Bank details will be handled externally by a trusted company (such as paypal), thus mitigating the threat of their compromise from my database. As well as this, all text inputs will

be validated and sanitised meaning the site has a lower risk of being attacked through malicious inputs.

## Regulatory guidelines and legal requirements:

The solution I intend to create must abide by the relevant legislation and guidelines for website development and green technology. In my research I found several pieces of legislation which fit the categories of my solution. I will explain which ones apply in my solution and why.

Environmental information regulations:
Since the company claims to be a green technology company, it could be necessary to provide environmental information related to their operations. This could be displayed as a page in my solution; however, it would use temporary test data as this information does not actually exist.

Greening Government ICT Strategy:
These are guidelines which involve moving ICT hosting infrastructures to collocated data centres. Although I have no control over this for my solution it would theoretically be beneficial to the company's green image.

WCAG 2.2:
The website must be made accessible according to WCAG guidelines.

UK GDPR:
All user data must be handled in accordance with GDPR

Equality Act:
The site cannot facilitate any forms of discrimination

DMCCA:
Cannot include drip pricing, misleading green claims or fake reviews. Misleading green claims is especially important in this case.

PECR:
Users must opt in for marketing if there is any, and cookies. Users must be informed as to which cookies are used and what for.

Copyright, Designs and Patents Act:
Any assets used my either be original, properly licensed, in the public domain or falling under fair use

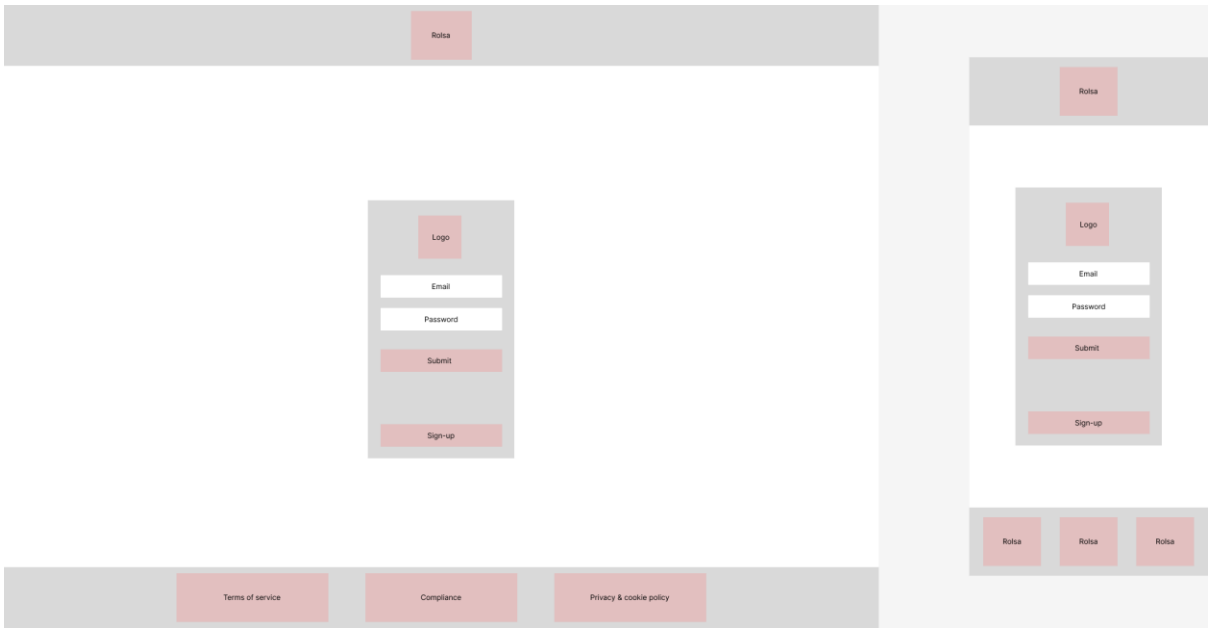
# Design Documentation:

## Visual interface diagrams:

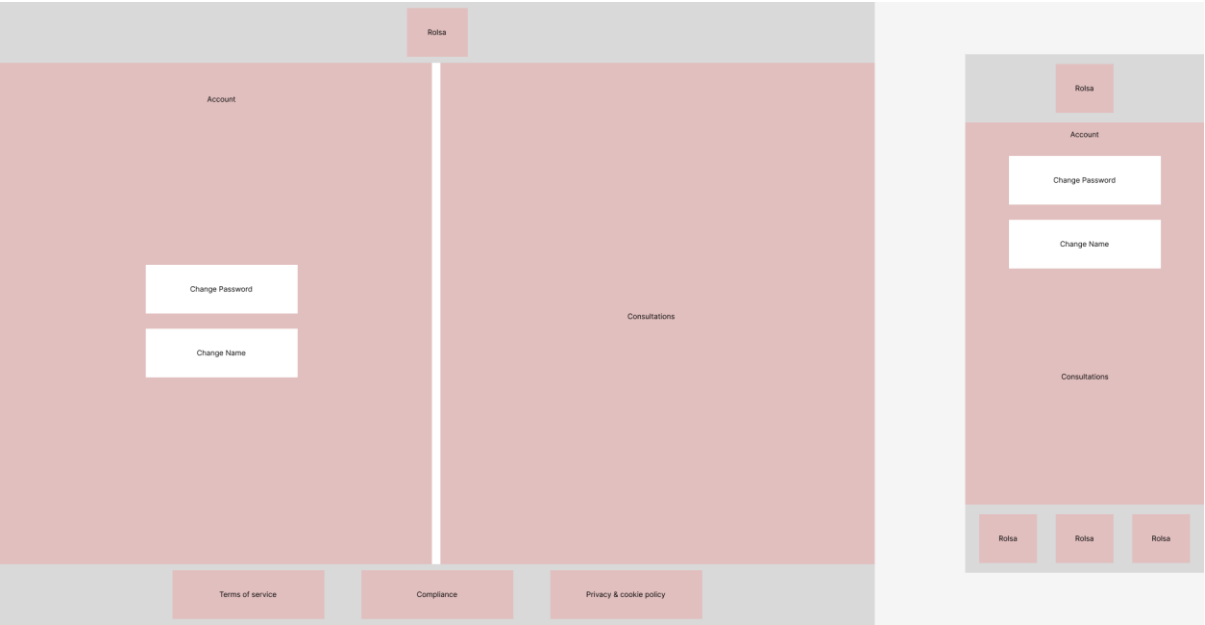
## Home page:

- When the user has logged in, the login and signup options will be replaced with a link to the accounts page

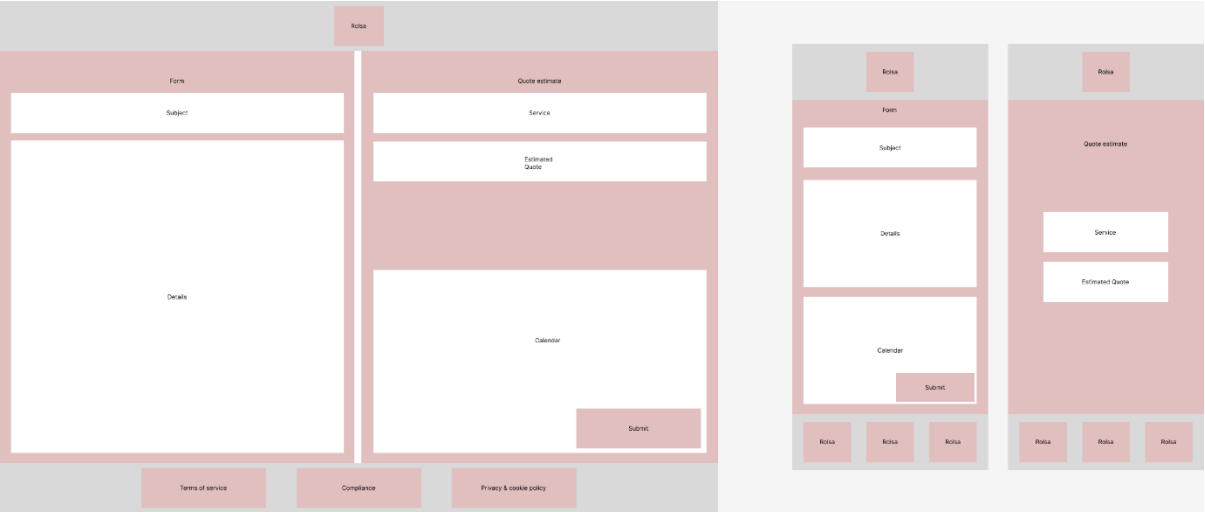
## Registration and login:




## Account and consultation management page:

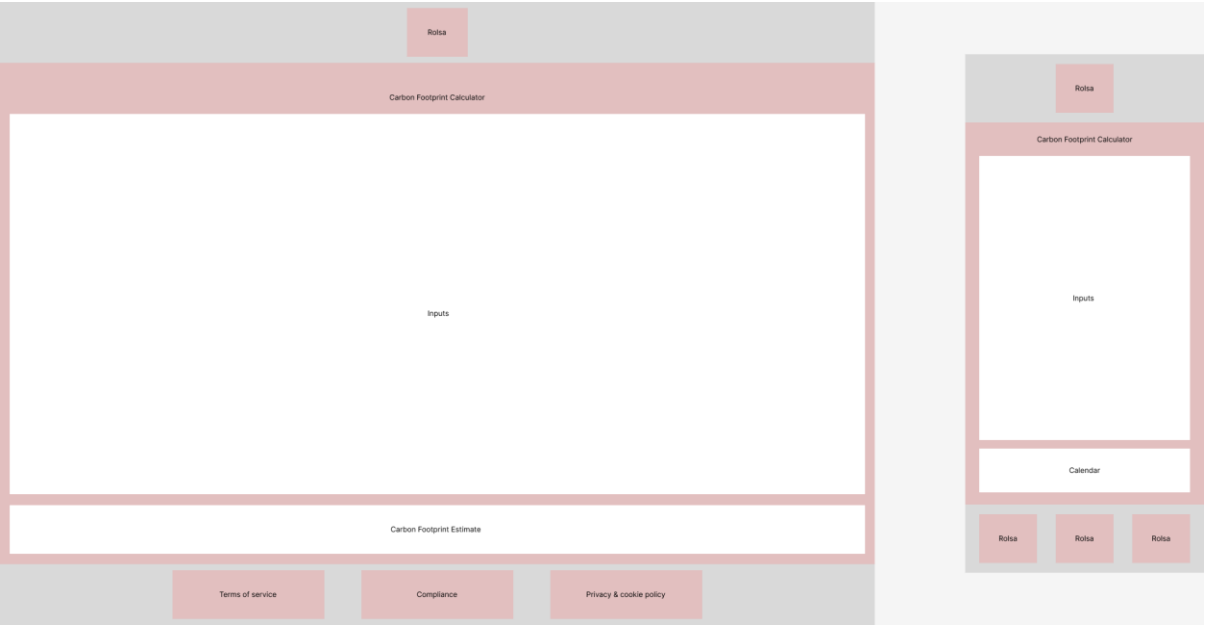



## Consultation booking page:

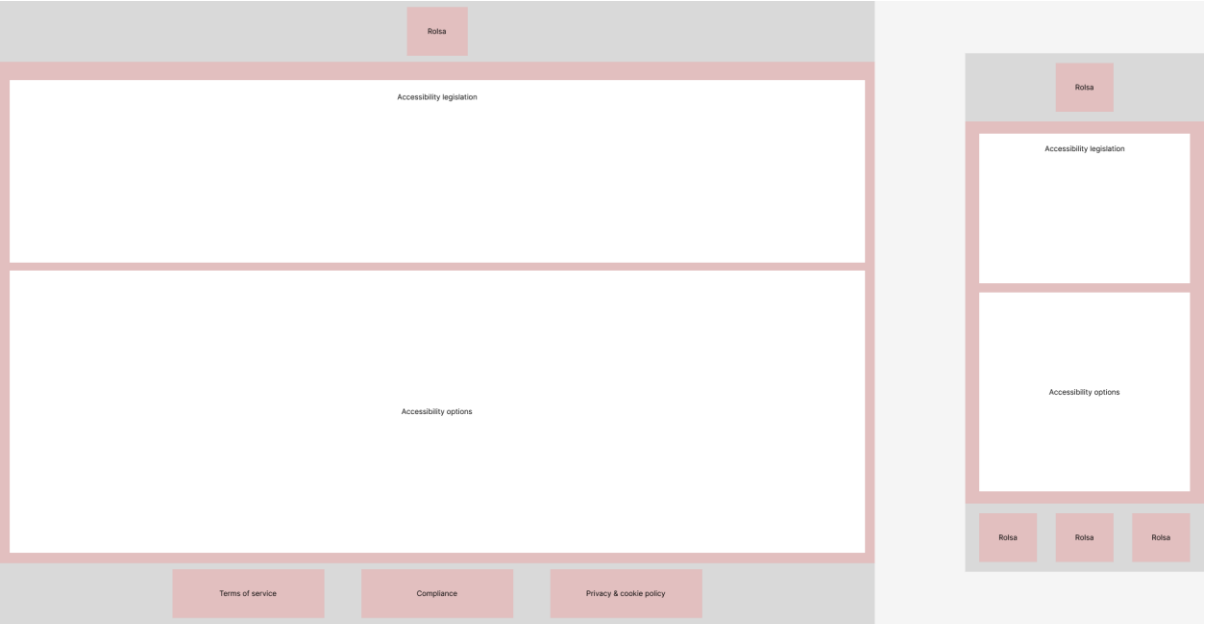



- Since there is a large amount of content on this page, on the mobile version, the user can scroll through the content to make the space less cramped

## Carbon footprint calculator page:

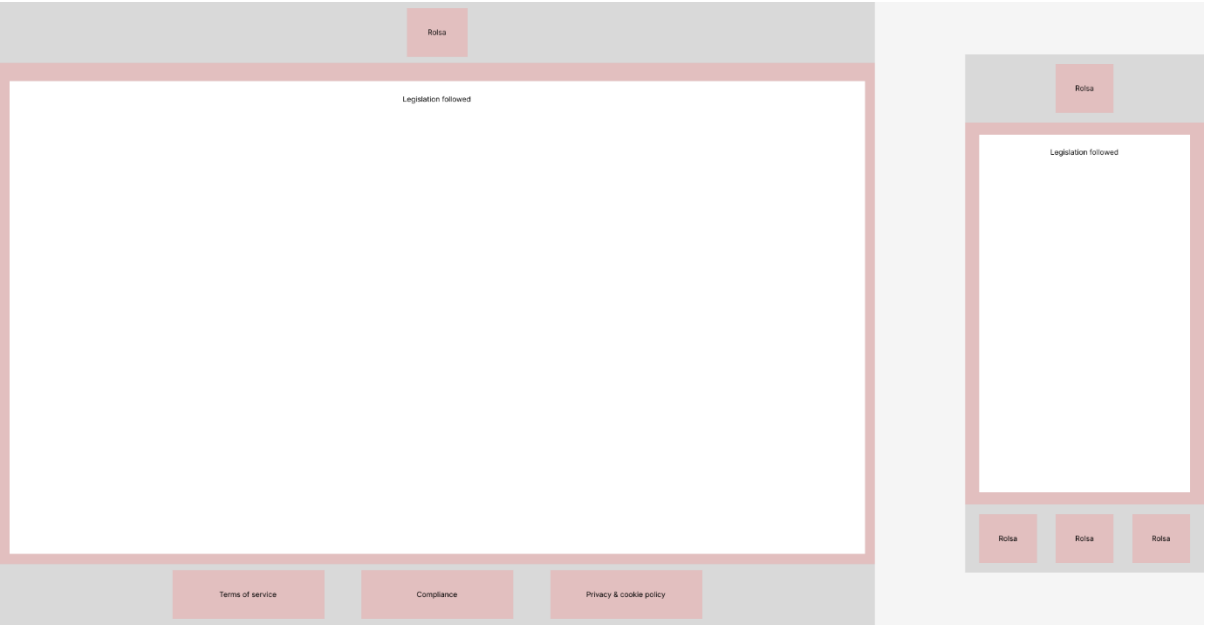



## Accessibility page:



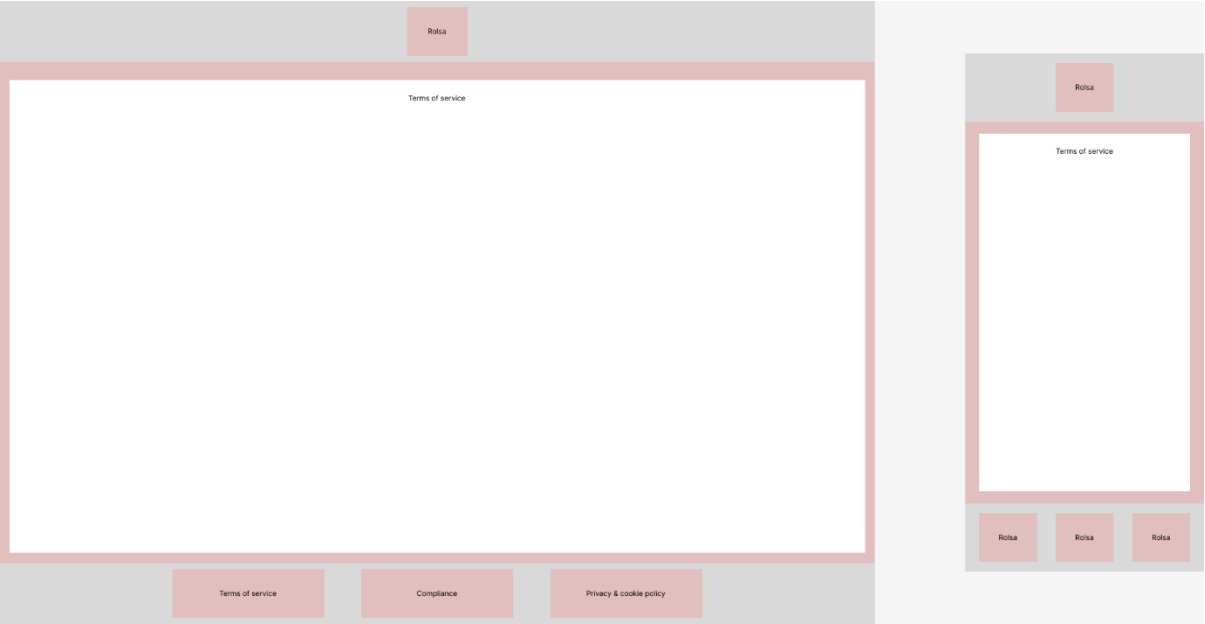

- The top half of this page will explain which accessibility guidelines are followed (WCAG) and how. The bottom half of this page will include options such as bold text, large text, changing the background colour to yellow where there is white. Other options are covered by browsers such as narrators; however, it is important that the HTML / PHP is implemented so that it is compatible with this

## Compliance page:




- This page will detail all the relevant guidelines and legislation which are followed, as detailed in the regulatory guidelines and legal requirements section

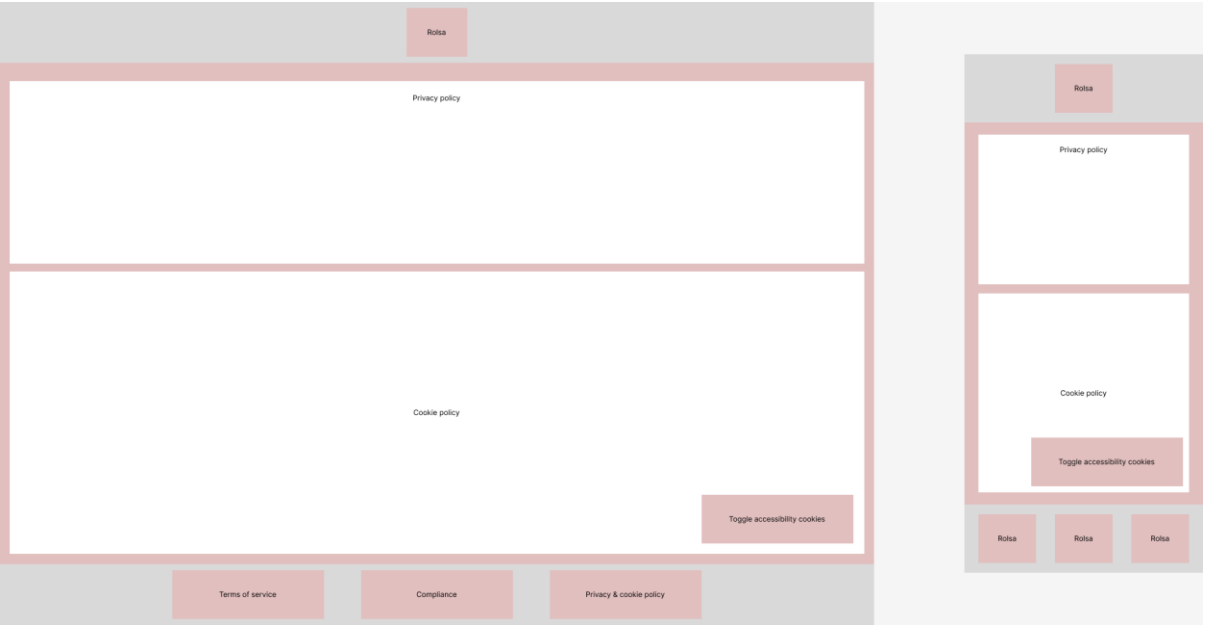
## Terms of service page:




- This page details the terms of service for using the website

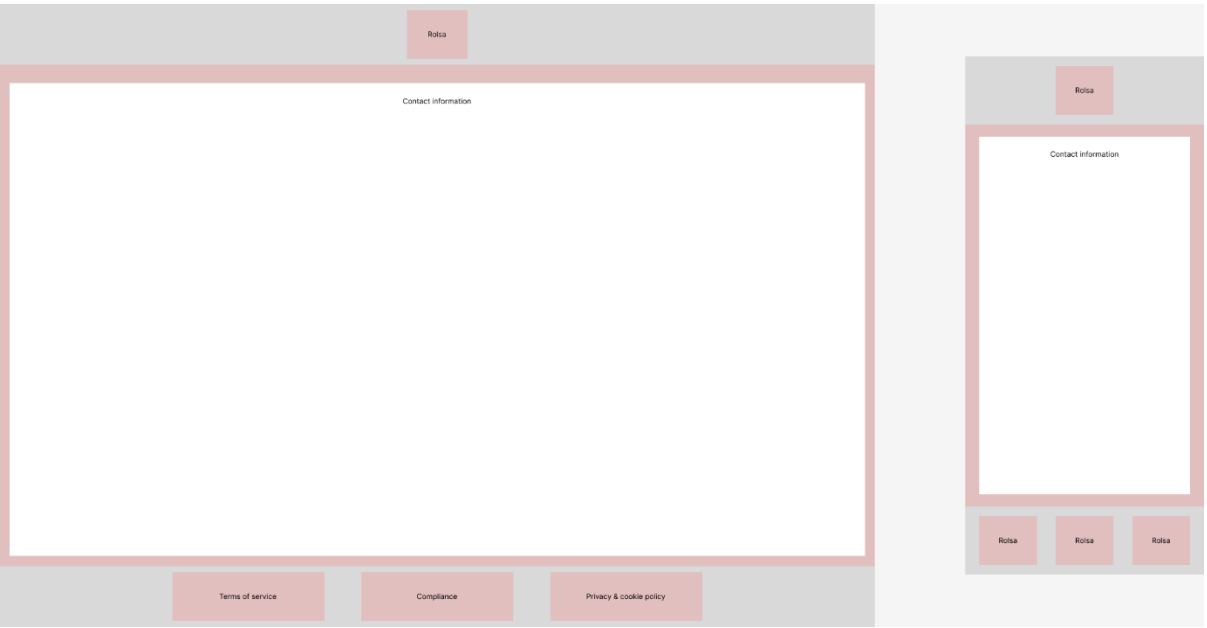
All pages will follow this format, including any additional pages should more be included in the final prototype.

## Privacy policy and cookie policy page:




- The privacy policy details how the website collects, uses and shares user data whereas the cookie policy details data gathered via cookies. Due to the data protection act, accessibility cookies can be turned off since they are not strictly necessary for the website's basic functionality, whereas login sessions are

## Contact page:

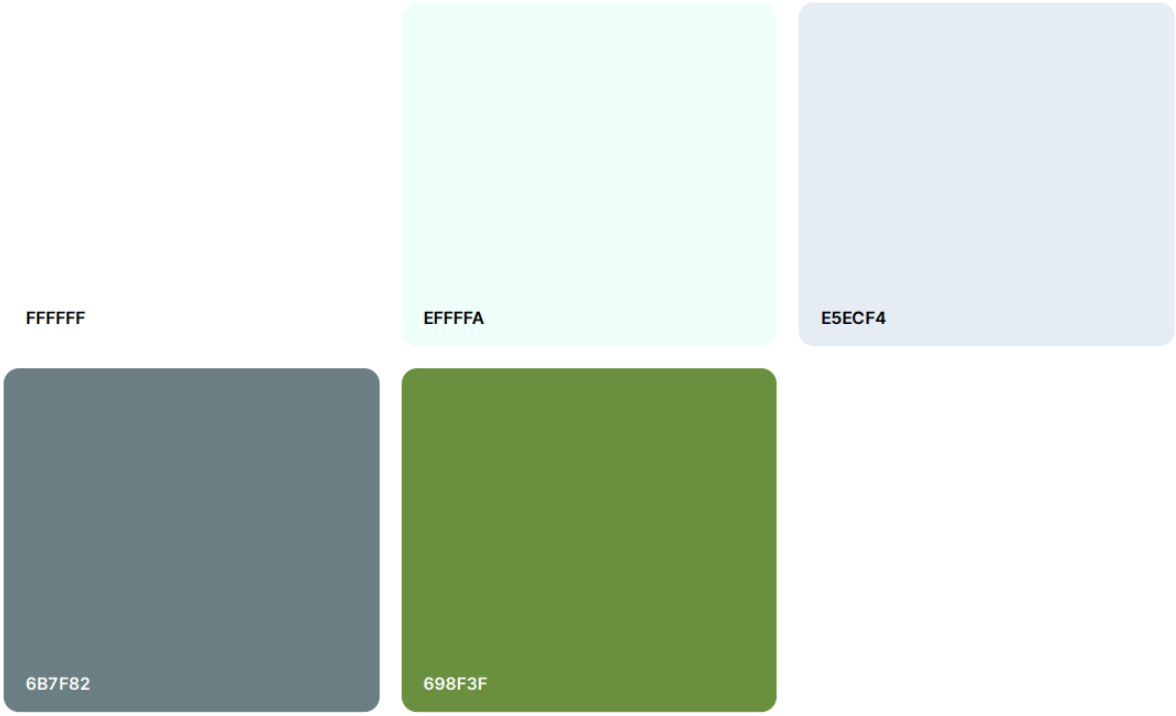


## Colour scheme:

Since the site focuses on green technology, I will user a colour scheme which feels futuristic and polished evoking senses of cleanliness and optimism to users. This will involve light blue and green shades with grey and white tones.

**Prototype colour scheme:**

Using a colour scheme generator, I created this early version of my intended scheme:




## Data requirements:

To create a solution which follows my proposal, there are many data requirements:

Users:
- user_id(pk)
- name
- email(unique)
- password_hash
- address

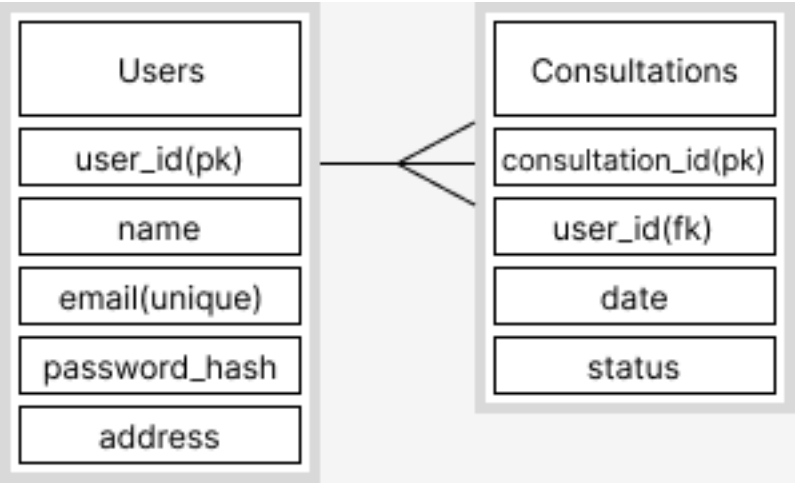
In this database, users have an auto incrementing primary key. This means that they are identified by number, which is naturally unique and cannot change, rather than by name. Email is set to unique to implement validation blocking users from registering the same email twice.

Consultations:
- consultation_id(pk)

- user_id(fk)
- date
- status

As this website will be a prototype, when a consultation is complete there will be a default quote amount which is pushed to the user. In a later version, a specialist would manually write this quote. The consultations database uses the user id as a foreign key allowing it to associate bookings with individuals. The status can be either upcoming, cancelled or completed – it will be stored as an ENUM. The date will be stored as a date type.
My databases will be programmed in SQL lite and will be secured to prevent SQL injections. The password_hash will use the PHP hashing function to do so, which also automatically salts the password, adding an extra random string for security.

## Entity relationship diagram:




This is my entity relationship diagram (ERD) for the relationship between users and consultations. Each user can have many consultations as a user can have several needs, whereas each consultation only needs one user. If the database included specialists, then each specialist would need to be assigned to a given number of consultations, however as this is a prototype, I chosen to leave out specialists for now. Consultations will not be removed from the database once they have commenced as after the consultation a user will be left with a quote. If the user choses to cancel their quote, the consultation will be removed from the database.

## Data dictionaries:

### Users:

| Fiend name | Data type | Key | Format |
|---|---|---|---|
| **user_id** | INT | PK | Auto increment |

| | | | |
|---|---|---|---|
| **name** | VARCHAR(255) | | No special characters, apostrophes accepted |
| **email(unique)** | VARCHAR(255) | Unique | " ***@***.*** " |
| **password_hash** | VARCHAR(255) | | At least one capital, one lowercase, a number and a special character. Must be 8 characters or more. |
| **address** | VARCHAR(255) | | House number, road, City, county, country, post code |

## Consultations:

| Field name | Data type | Key | Format |
|---|---|---|---|
| **consultation_id** | INT | PK | Auto increment |
| **user_id** | INT | FK | Auto increment |
| **date** | DATETIME | | YYYY-MM-DD, HH:MI:SS |
| **status** | ENUM | | 'upcoming', 'cancelled', 'completed' |

## Session:

| Field name | Data type | Key | Format |
|---|---|---|---|
| **session_id** | INT | PK | Auto increment |
| **user_id** | INT | FK | Auto increment |

## Test strategy:

The objective of the test strategy is to ensure that all non-functional and functional requirements are satisfied in accordance with the user acceptance criteria and key performance indicators.

**Home page:**

| Test description | Inputs | Expected outputs | Comments |
|---|---|---|---|
| **Navigation** | Click links to all other pages from home page | Scrolling functions. Links function and | |

| | | | |
|---|---|---|---|
| | | redirect the user to their respective pages | |
| **Information access** | | Content displayed is meaningful and legible to an arbitrary satisfactory degree. User can access information about products and services on the market | |
| **Accessibility** | | Browser accessibility options are compatible with the site and accessibility page options are functional | |

**Account registration:**

| Test description | Inputs | Expected outputs | Comments |
|---|---|---|---|
| **Navigation** | Click home page link | Scrolling functions. Home page link correctly redirects user to home page | |
| **Account creation - email** | Email in wrong format, email in correct format, password and confirm password same and in correct format, name | Account creation should not commence if the user enters an email in an incorrect format | |
| **Account creation - password** | Email in correct format, password in wrong format, password in correct format, password in correct format with non-matching | Account creation should not commence unless the user enters two equal passwords each in the correct format | |

| | | | |
|---|---|---|---|
| | confirmation password | | |
| **Account creation - name** | Valid name, empty string | Account should only be created if the name is valid | |
| **Account creation – SQL injection** | SQL injection attempt | All inputs should be sanitised, preventing an SQL injection | |
| **Login - email** | Email not in database, email in database, correct password | Login should not commence unless the email exists in the database | |
| **Login - password** | Correct password, incorrect password, password for another user | Login should not commence unless the password is correct and associated with the inputted email | |
| **Login – SQL injection** | SQL injection attempt | All inputs should be sanitised, preventing an SQL injection | |
| **Change password - email** | Existing email, non-existing email | Emails which are not found in the database should raise an error and show a link to the account creation page, emails in the database should give the user an input for a new password | In a final version, an email linking the password input would be sent to the user |
| **Change password - password** | Password in wrong format, password in correct format, password in correct format with non-matching confirmation password | Password should only be changed if in the correct format with a matching confirmation password | |
| **Change password – SQL injection** | SQL injection attempt | All inputs should be sanitised, preventing an SQL injection | |

**Consultation management and account management page:**

| Test description | Inputs | Expected outputs | Comments |
|---|---|---|---|
| **Navigation** | Click home page link | Scrolling functions. Links function and redirect the user to their respective pages | |
| **Change name** | Empty string, valid name | User should only be able to change their name to one of a valid format | |
| **Change name – SQL injection** | SQL injection attempt | All inputs should be sanitised, preventing an SQL injection | |
| **Pay quote** | Click payment button | Quote should be listed as paid after clicking payment button | Paypal sandbox program could be implemented for more realistic payment process |
| **Pay for consultation** | Click payment button | Consultation should be listed as paid after clicking payment button | Paypal sandbox program could be implemented for more realistic payment process |
| **Cancel consultation** | Click cancel button | Consultation should disappear after clicking cancel button | Paypal sandbox program could be implemented for more realistic payment process |
| **Reject quote** | Click cancel button | Consultation should disappear after clicking cancel button | Paypal sandbox program could be implemented for more realistic payment process |

**Consultation booking page:**

| Test description | Inputs | Expected outputs | Comments |
|---|---|---|---|
| **Navigation** | Click home page link | Scrolling functions. Links function and redirect the user to their respective pages | |
| **Form** | Submit button, form contents | User can submit the form | |

| | | | |
|---|---|---|---|
| **Form – SQL injection** | SQL injection attempt | All inputs should be sanitised, preventing an SQL injection | |
| **Calendar** | Click on an available date and an unavailable date | User should be given a consultation if they click an available date and an error message if there is an unavailable date | The program could also include specific times |
| **Calendar – SQL injection** | SQL injection attempt | All inputs should be sanitised, preventing an SQL injection | |
| **Estimated quote generator** | Service, unknown service, amount | The user should get an estimated quote only when inputting a service listed on the form and an amount e.g. 30 solar panels installed | |
| **Estimated quote generator – SQL injection** | SQL injection attempt | All inputs should be sanitised, preventing an SQL injection | |

**Carbon footprint calculator page:**

| Test description | Inputs | Expected outputs | Comments |
|---|---|---|---|
| **Navigation** | Click home page link | Scrolling functions. Links function and redirect the user to their respective pages | |
| **Calculator** | Sources of carbon inputted via form | User should get an estimate for their footprint and personalised advice on how to reduce it based on their personal biggest carbon source | |
| **Calculator – SQL injection** | SQL injection attempt | All inputs should be sanitised, | |

| | | | |
|---|---|---|---|
| | | preventing an SQL injection | |

**Accessibility page:**

| Test description | Inputs | Expected outputs | Comments |
|---|---|---|---|
| **Navigation** | Click home page link | Scrolling functions. Links function and redirect the user to their respective pages | |
| **Information access** | | Content displayed is meaningful and legible to an arbitrary satisfactory degree. User can access information about how the site follows WCAG Guidelines | |
| **Turn on accessibility features** | Toggle accessibility features switch | Text should be larger and bold | |

**Compliance page:**

| Test description | Inputs | Expected outputs | Comments |
|---|---|---|---|
| **Navigation** | Click home page link | Scrolling functions. Links function and redirect the user to their respective pages | |
| **Information access** | | Content displayed is meaningful and legible to an arbitrary satisfactory degree. User can access information about how the site follows legislation and guidelines | |

**Terms of service page:**

| Test description | Inputs | Expected outputs | Comments |
|---|---|---|---|
| **Navigation** | Click home page link | Scrolling functions. Links function and redirect the user to their respective pages | |
| **Information access** | | Content displayed is meaningful and legible to an arbitrary satisfactory degree. User can access information about the site's terms of service | |

**Privacy policy and cookie policy:**

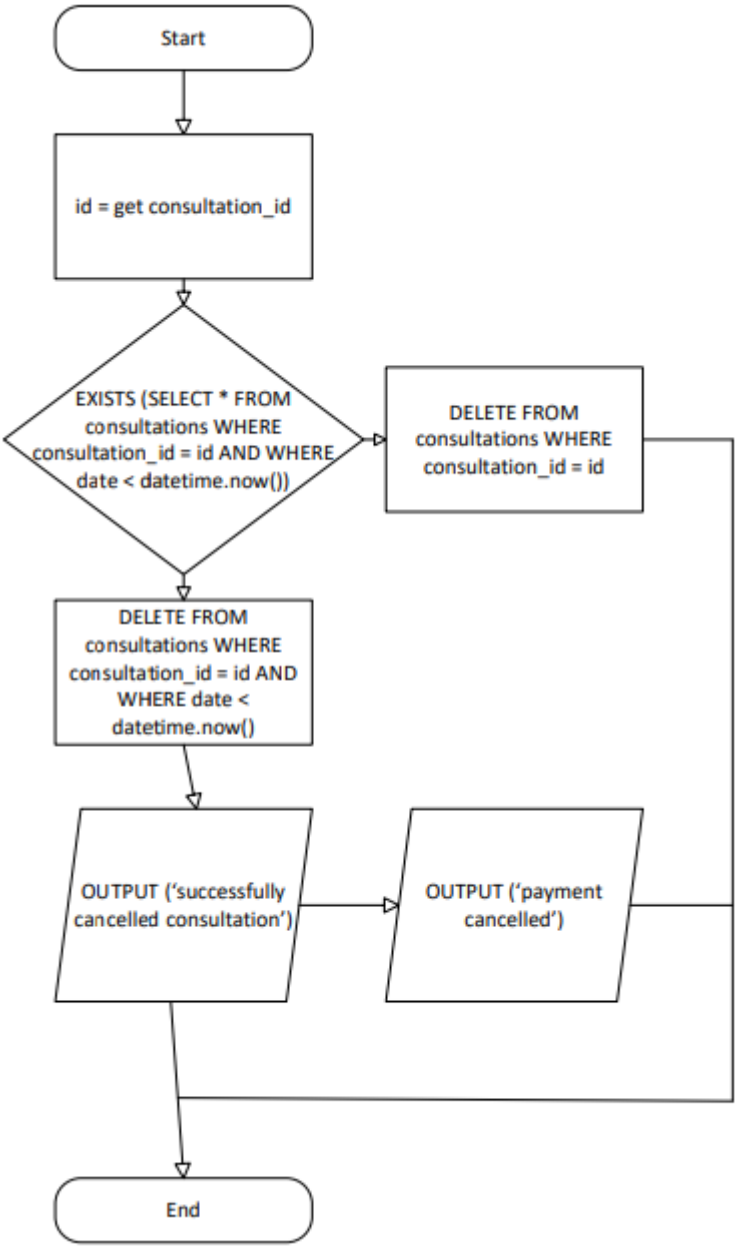
| Test description | Inputs | Expected outputs | Comments |
|---|---|---|---|
| **Navigation** | Click home page link | Scrolling functions. Links function and redirect the user to their respective pages | |
| **Information access** | | Content displayed is meaningful and legible to an arbitrary satisfactory degree. User can access information about the site's privacy policy and cookie policy | |
| **Turn off accessibility cookies** | Click toggle for accessibility cookies | All accessibility cookies should be terminated; new ones should not be created unless this is un toggled | |

**Contact page:**

| Test description | Inputs | Expected outputs | Comments |
|---|---|---|---|
| **Navigation** | Click home page link | Scrolling functions. Links function and redirect the user to their respective pages | |
| **Information access** | | Content displayed is meaningful and legible to an arbitrary satisfactory degree. User can access contact information | |

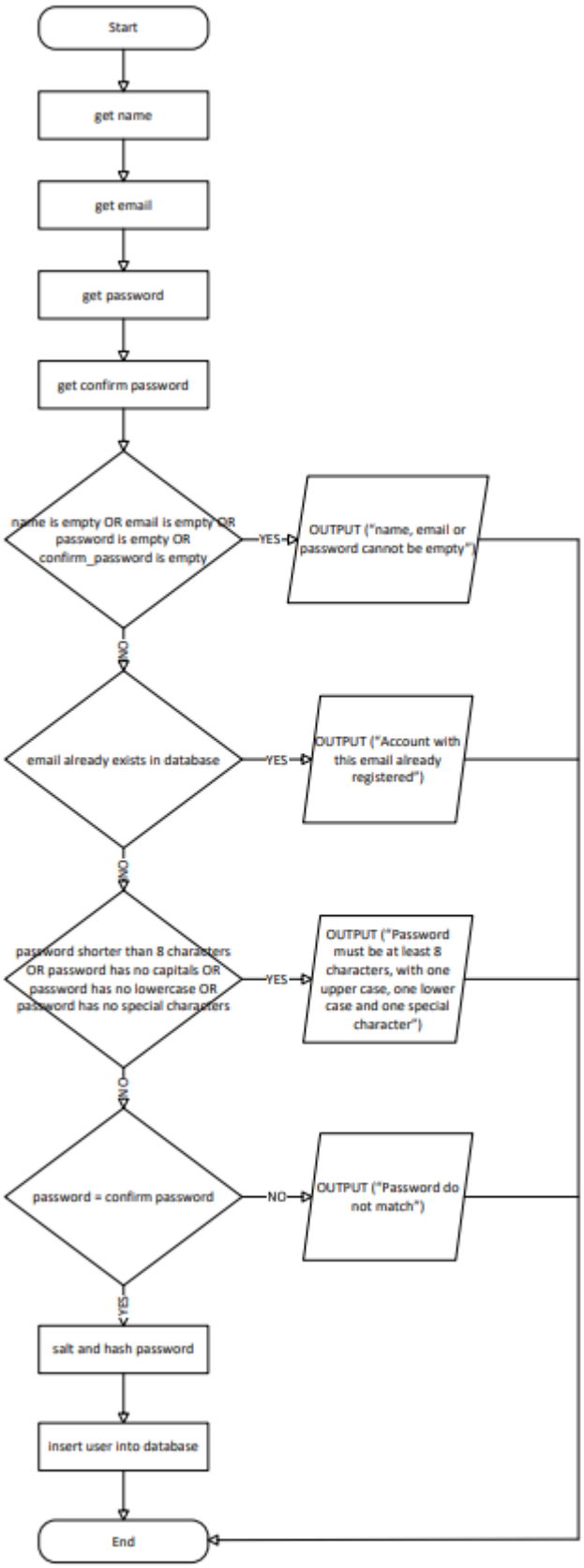
# Algorithm designs:

## Cancel consultation / payment:



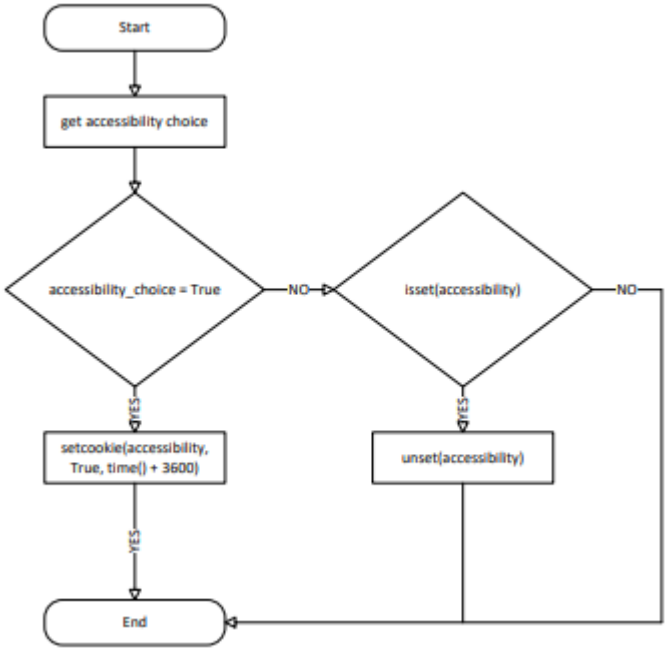

This is a high-level simplified version of a cancellation algorithm. I have chosen to design it so that when consultations taking place after the current time are cancelled, it is considered a cancelled payment for a quote or a rejection of it. This program would run when a user selects a cancel button on the account and consultation management page.

## Sign-up:



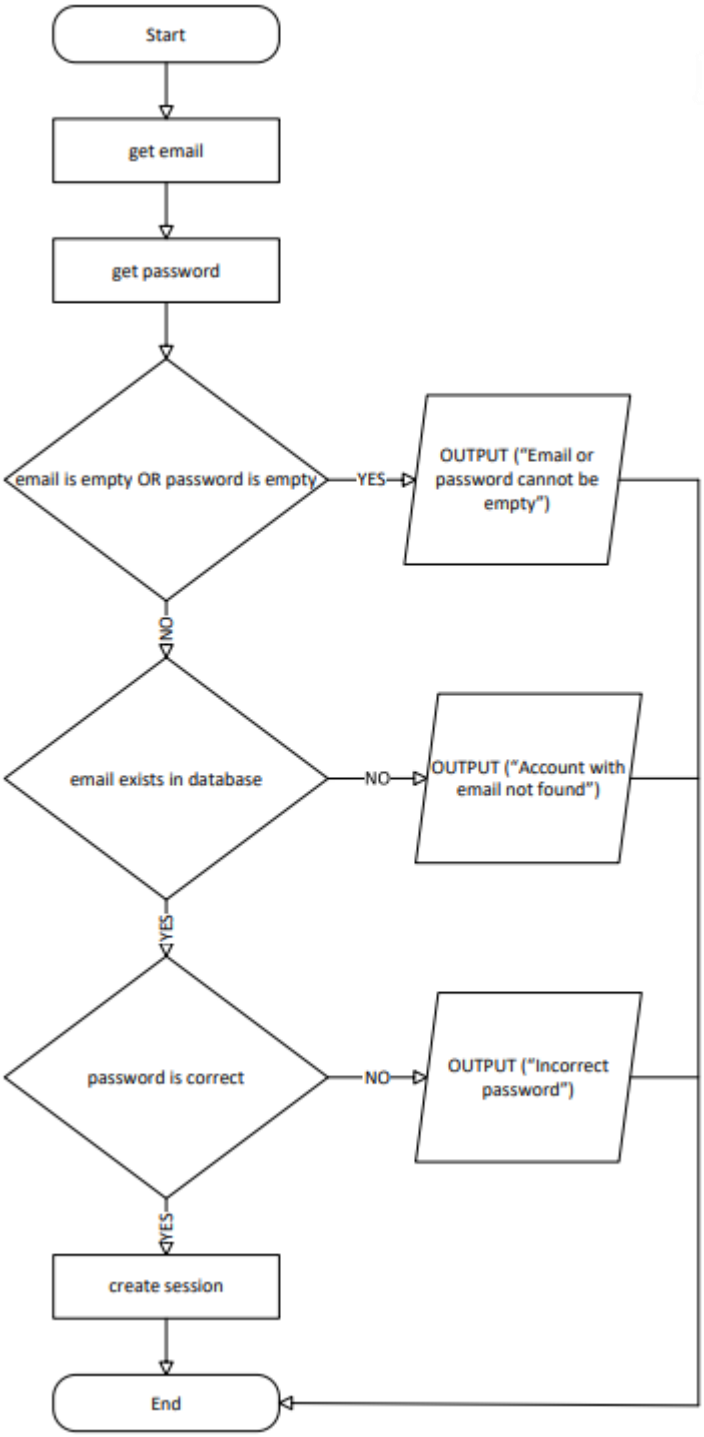

This algorithm creates a new user in the database, under the condition that the passwords are the same and the email does not exist in the database, along with other format conditions. In an actual program, PHPs built in formatting functions would also be utilised to format emails correctly.

## Set accessibility cookie:



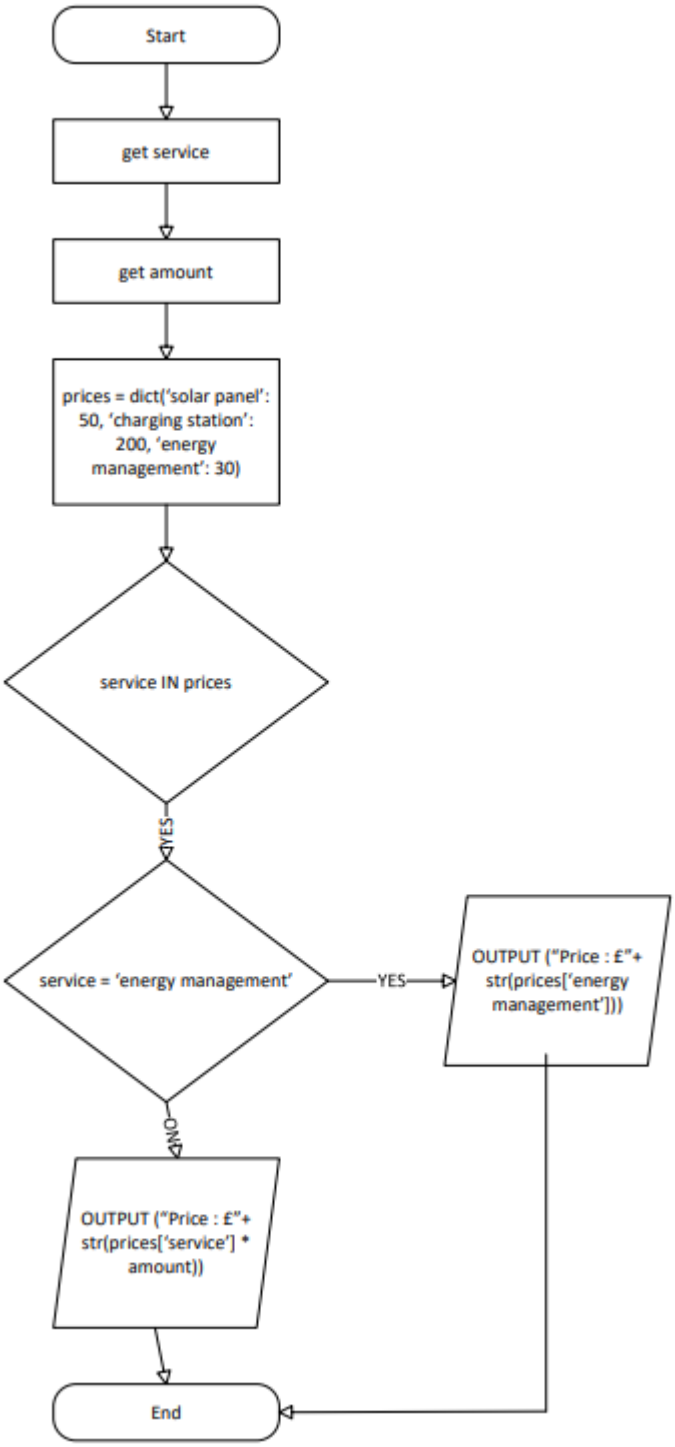

This program will set a cookie to be used to alter the appearance and other settings on the website. If the cookie is set and then turned off, it will be deleted. By default, it does not exist.

## Login:




This program logs the user in if they enter the correct email and password. Logins are handled through sessions in this program.

## Estimated quote:




This program gives the user an estimate of the cost of their needed services. I have made it such that the smart home energy management systems are a one time £30 payment since a user would not purchase more than one system. I have also estimated rough prices, since this is for a prototype the exact prices of services could be implemented later.

# Research

## Hardware and software within the context of the industry:

Rosla Technologies are a green technology company, therefore the proposal and solution should make use of technology frequently successful in this industry.

### Software used within the green technology industry:

Carbon accounting and reporting software:

"Carbon accounting and reporting software allows companies to calculate their business's carbon footprint and understand where emissions come from "

- *Carbon Accounting explained | normative*. Available at: https://normative.io/insight/carbon-accounting-explained/ (Accessed: 05 January 2026).

Regulated markets such as the EU and UK are mandated or compelled to use carbon accounting software. This comes from the EU legislation CSRD (Corporate Sustainability Reporting Directive), which affects around 50,000 companies. The UK version of this legislation is SECR (Streamlined Energy and Carbon Reporting).

While this software is largely optional, it would be beneficial for a company advocating green energy to make use of them to gain insight for public distribution of information about company achievements, with respect to environmental sustainability.

Carbon accounting and reporting software include:

- Salesforce Net Zero Cloud
- Microsoft Sustainability Cloud
- Sphera
- Greenly

Life cycle assessment (LCA) tools:

For a company which offers services and products, it is important to evaluate both the life span and environmental impact the creation of a product has. Life cycle assessment tools provide a structured framework which can quantify a service or product's entire life span and environmental footprint. The analysis of which can then be used to identify hot spots of environmental harm and work on these weaknesses.

Examples of some life cycle assessment tools are:

- SimaPro
- Sphera (GaBi)
- OpenLCA

IoT & Smart Environments:

“Smart environments aim to satisfy the experience of individuals from every environment, by replacing hazardous work, physical labour and, repetitive tasks with automated agents”

- *Smart environment* (2024) *Wikipedia*. Available at: https://en.wikipedia.org/wiki/Smart_environment (Accessed: 05 January 2026).

In green technology, smart environments use IoT sensors, AI and data analytics to create more efficient systems in order to optimise energy usage and reduce waste.

IoT based logistic software are also used by green technology companies to streamline stock control and boost operational efficiency. For example, keeping track of solar panels in stock.

Websites:

Green technology companies use websites for a wide range of reasons. These could be advertisements, displaying commitment to the cause of sustainability, displaying an eco-score. Most websites hosted by green technology companies will be used for both product and service advertising and the effective portrayal of a company ethos to both employees and the public.

Websites will also be used of an employee basis in order to process orders, payments and other sections of their jobs.

Some examples of green technology websites will be attached as images and links at the end of this section of the document

Business Intelligence (BI) software:

BI software helps organisations analyse data, to create reports and gain insights about company operations. It is standard for a large variety of companies to make use of BI software.

Business intelligence software includes:

- Microsoft Power BI
- Domo
- Mode Analytics
- Qlik

## Hardware used within the green technology industry:

Energy Efficient components:

Networking and computer clients are chosen in an optimised fashion for a balanced compromise between efficiency and energy consumption which prioritises carbon efficiency. For example, thin client models are often utilised to reduce consumption on a per machine basis.

These devices are also frequently energy certified to align with company ethos.

IoT devices:

Sensors allow real time monitoring of data produced by company services and products. For example, a green technology company could produce smart monitors, which allow users to control heating from a mobile device. Thermal sensors can be used to display the current room temperature to these users on their device.

Smart labels allow efficient logistic operations, feeding data to business intelligence software in order to allow the company to expedite the process of buying new stock and tracking which stock is plentiful.

Hardware tools:

Since the company offers services of maintenance, they must have the capacity to do this and as such need a stock of hardware tools to work installation and maintenance.

Automotive hardware is used by companies to transport equipment and services. For a green technologies company, these processes are very important to optimise to reduce carbon emissions potentially by making use of electric vehicles.

## Newly emerging technologies:

Small Modular Reactors (SMRs) are advanced nuclear reactors which have a capacity around one third that of a traditional reactor. Producing this energy could be a future aspiration of a company such as Rolsa as well as working to distribute it and potentially themselves using it for greener company processes.

Next gen renewables are "Emerging solar panels made from materials known as perovskites [which] promise to be thinner, lighter and more flexible than traditional silicon"

- (No date a) *Next-generation solar panels redefine where clean energy can go - the renewable energy institute*. Available at: https://www.renewableinstitute.org/next-generation-solar-panels-redefine-where-clean-energy-can-go/ (Accessed: 05 January 2026).

These newly emerging solar panels will be important for a company like Rolsa to get involved in to keep adapting and improving their operations.

Long Duration Energy Storage (LDES) stores electricity for extended periods to balance the variability of renewable energy, making it much more reliable. This is a very important technology to green technology companies.

AI and big data are used to optimise energy efficiency and enhance resource usage.

## How digital solutions could be used to meet different user needs:

There are many different types of potential users including:
Customers:
- Businesses and Corporations
- Homeowners and individuals
- Public Sector Agencies

Investment community:
- Venture Capitalists
- Shareholders
- Financial Analysts

Talent pool:
- Job Seekers
- Potential Partners
- Researchers

General public:
- Media
- Students and Teachers

Additional users also exist such as people with various accessibility needs. Each of these users have various needs which could be met via digital solutions.

| User | Needs | Digital Solution |
|---|---|---|
| Businesses and Corporations | Integrated services, affiliate programs, APIs | A digital solution could facilitate joint ventures such as a green technology company providing solar panels at a discounted price to a research facility. Affiliate programs could be implemented such as linking to pages from which they source their hardware tools. The website could have an associated API which gives businesses data about various company metrics which could be useful |
| Homeowners and individuals | Information, contact and the ability to purchase services | Information pages should be made on the website displaying metrics and performance of technologies offered such as vehicle charging stations, as well as |

| | | |
|---|---|---|
| | | general guidance as to the function of services and products such as a smart home management system. A contact page must also be included. The page for purchasing would ideally need a database driven login and signup page with search functionality and the ability to buy products. |
| Public sector agencies | Information, contact and policy alignment with evidence of data protection | There should be information pages to display the purpose of the website and what the company offer as well as a contact page. The website should also contain a page which demonstrates their alignment with legislation. |
| Venture capitalists | Information and traction metrics | The website should show what problems this company solve and its popularity such that venture capitalists can decide on whether it is a worthy investment. This could be done through an order history page responsive to which product is selected. |
| Shareholders | Strategic updates and information, contact | 'About us' section can provide details on the strategic updates, company address and contact information should be made available. |
| Financial analysts | Financial information, contact and popularity metrics | Financial analysts should be able to see the popularity of products and services and should also be able to contact the company to obtain more detailed specific information. This could be through its own |

| | | |
|---|---|---|
| | | separate financial analyst request form. |
| Job seekers | Contact and information | Information on both the company's operations and ethos as well as job availability should be available (although jobs may be posted externally on sites such as linkedin). Contact should also be given and a form for potential job seekers could be provided on the website. |
| Potential partners | Information, contact and affiliate programs | Potential partners should be able to access information regarding the business and contact them as well as view any potential affiliate programs. This could be done through information sites, and an affiliate site. |
| Researchers and media | Information, sales metrics and contact | A contact page would be included for specific needed information. Many other information pages would also be included as well as visible sales metrics for products and services. |
| Students and teachers | Information and sales metrics | Information must be made accessible on the website potentially without logging in as well as sales metrics being visible. |
| Accessibility | The site must be easy to use for all with no inadvertent ableism | The site must follow Web Content Accessibility Guidelines (WCAG) 2.2 (or the most recent version) |

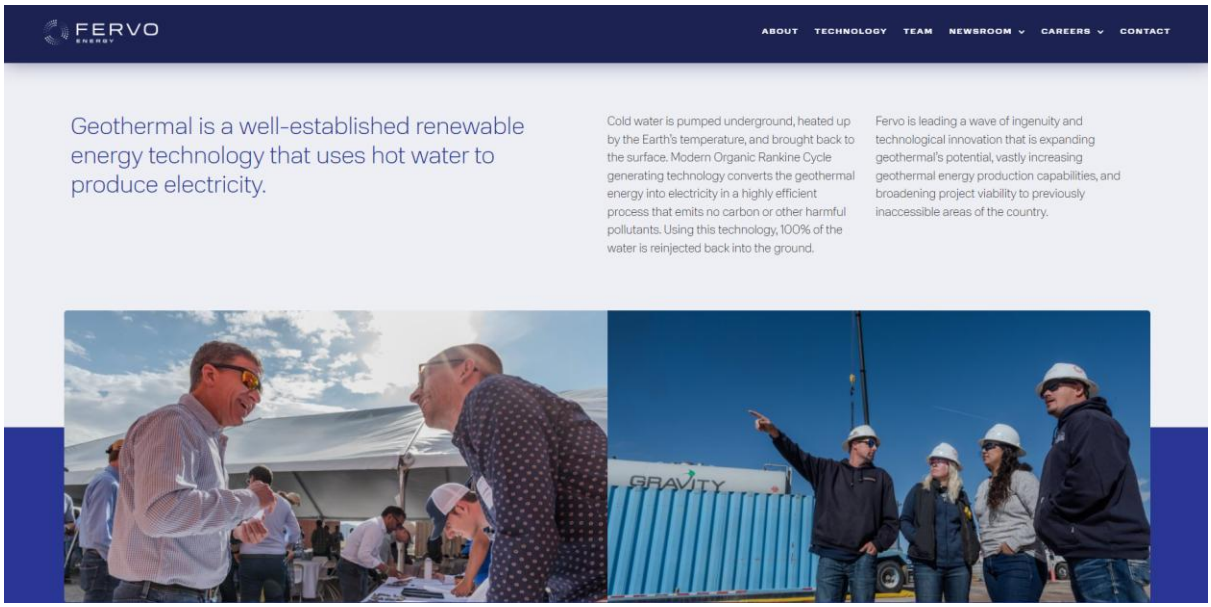
## How existing solutions meet these needs:

Many preexisting solutions meet these needs; I will document a few which do and how they achieve this.
Companies I will cover are:

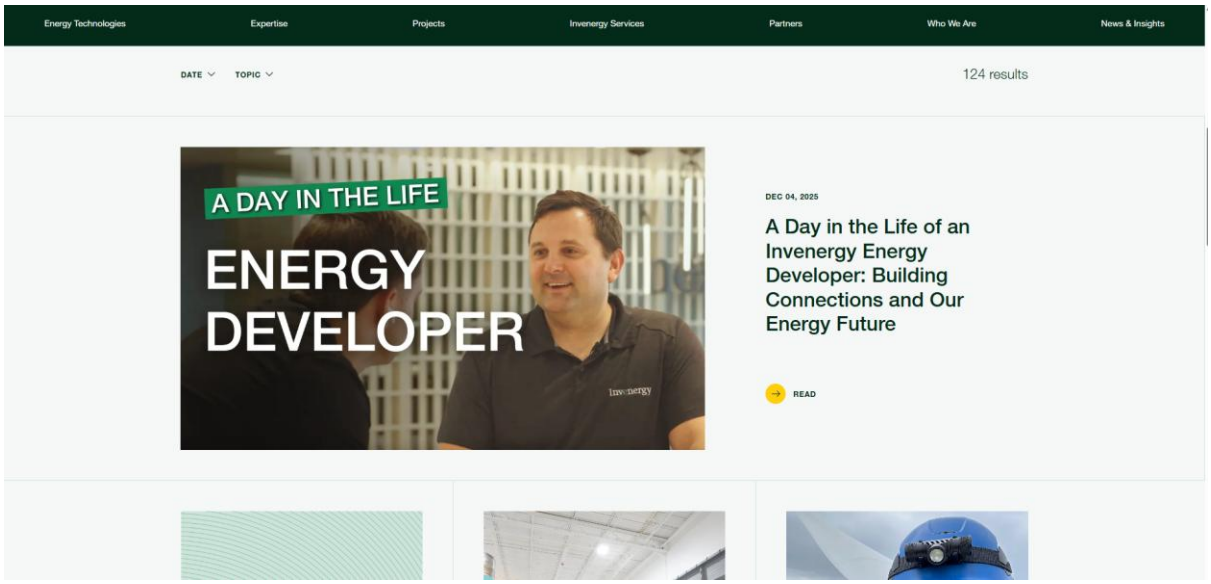
- Fervo Energy - https://fervoenergy.com/

- Invenergy - https://invenergy.com/
- Heliogen - https://www.heliogen.com/
- Carbon Clean - https://www.carbonclean.com/

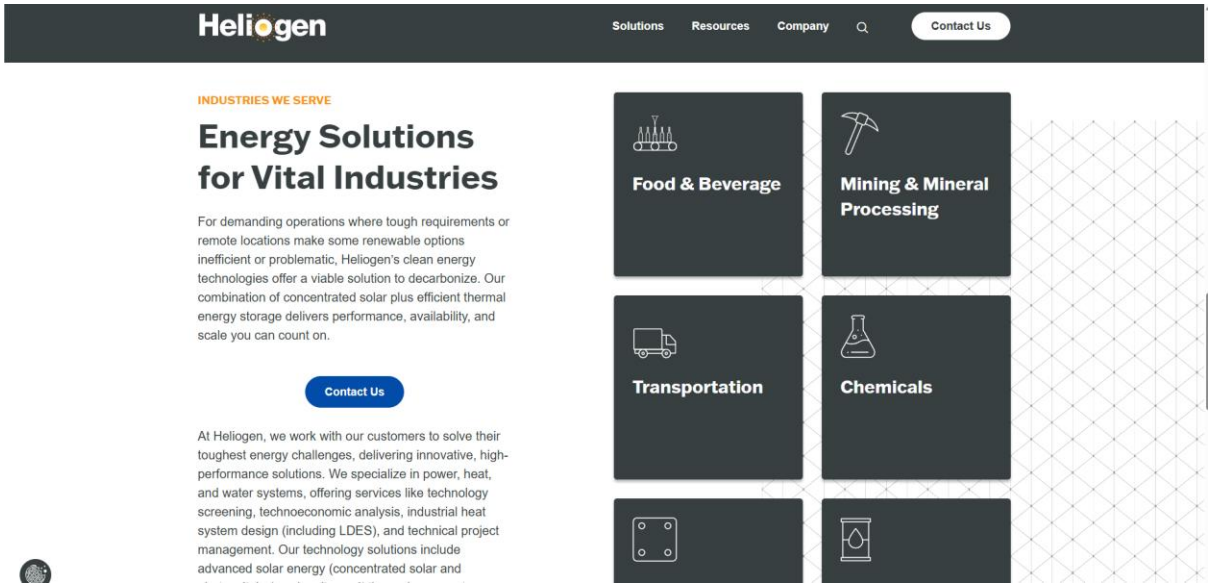
Information:



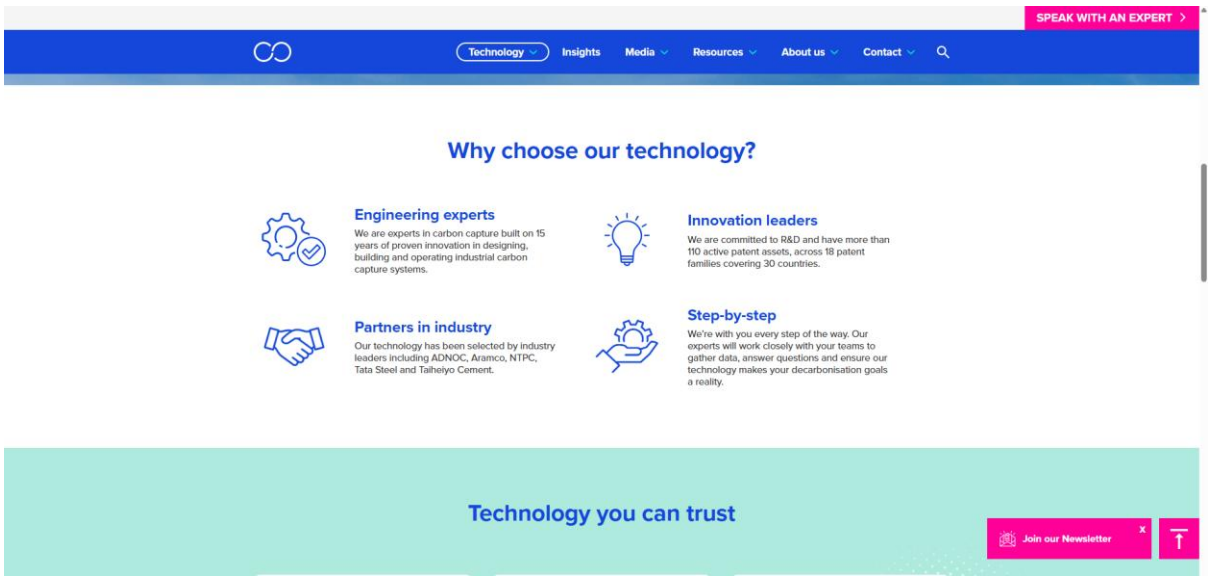

- From the fervo energy home page, the about button is clearly visible in the page header, which directs you to a page filled with information




- Invenergy has a 'latest news' page which delivers up to date information about company operations. As well as this, on the home page, scrolling down affords the user various other company information




- Much like Invenergy's page, heliogen also has a newsletter site and scrolling information, however it also contains specific detail on energy solutions

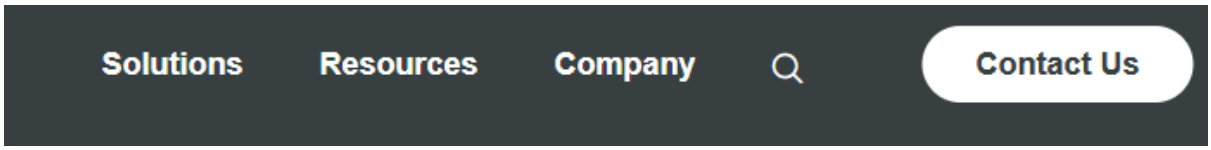



- The Carbon Clean website features a home page with a link to information immediately in the centre entitled 'find out more'

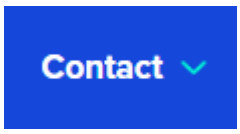
Contact:

- Fervo has a contact link immediately visible on the home page header

- Invenergy offered a contact us link collapsed under the news and insights section of the header

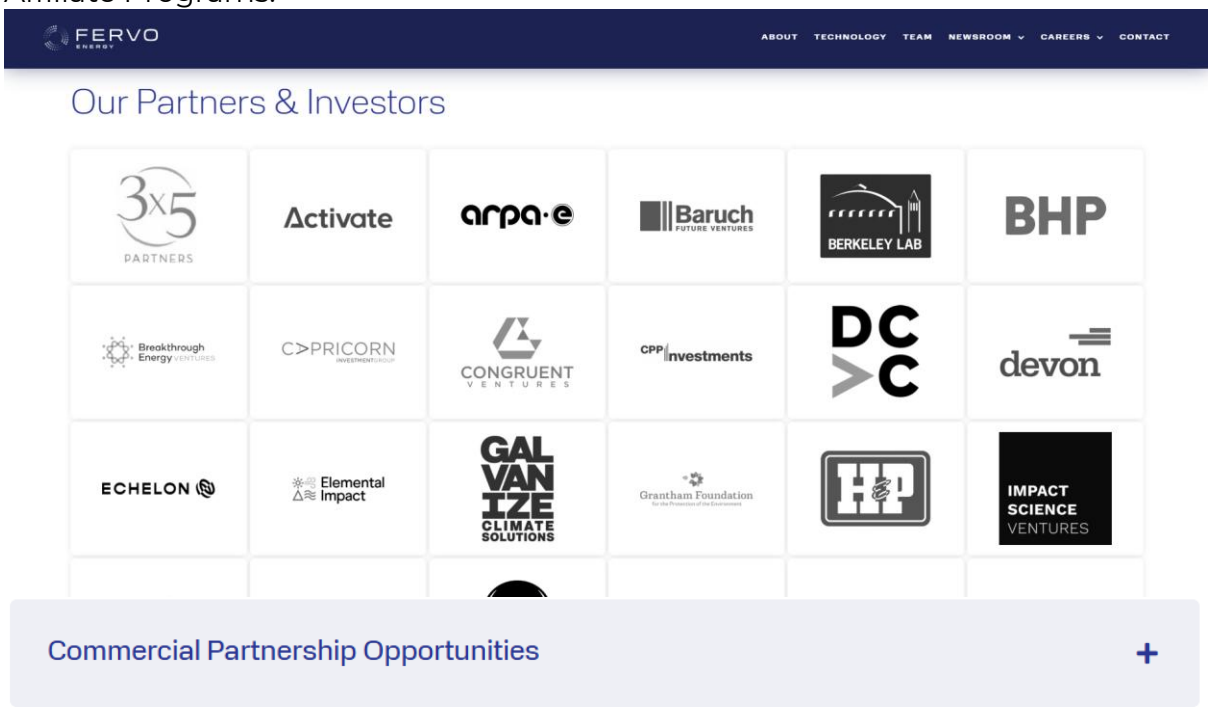



- Heliogen used a contact us link in the main header

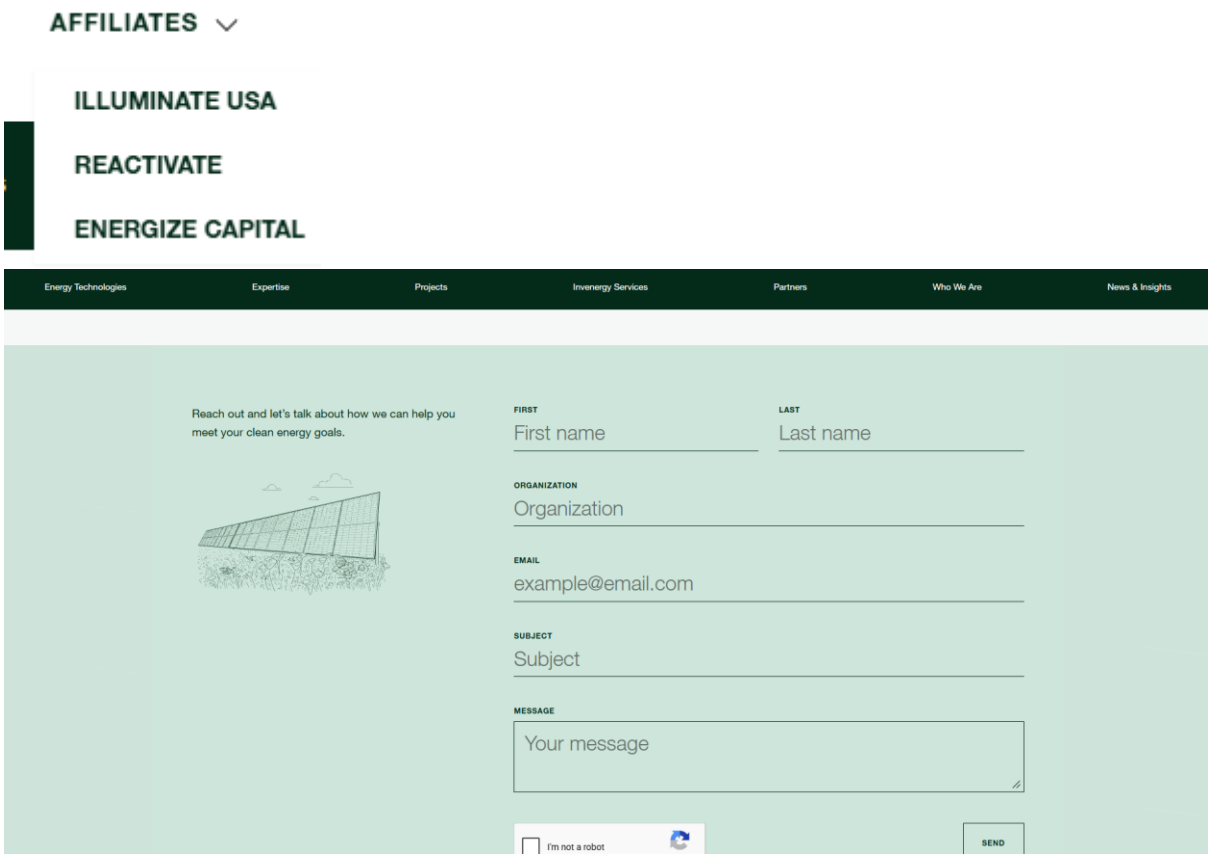



- Carbon clean offered a collapsed contact button in the header which gives two options, sign up for the newsletter or speak with an expert

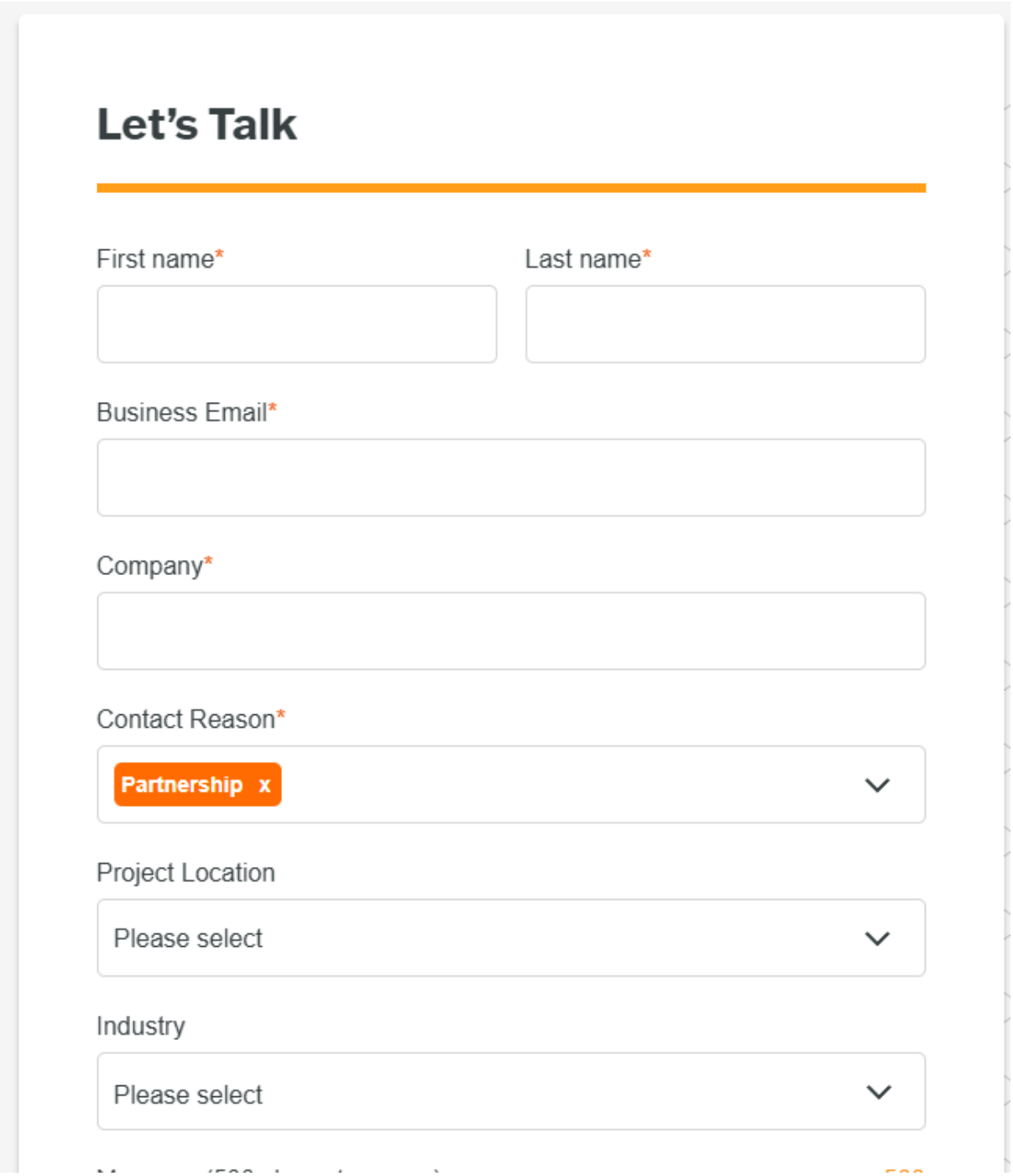
Affiliate Programs:



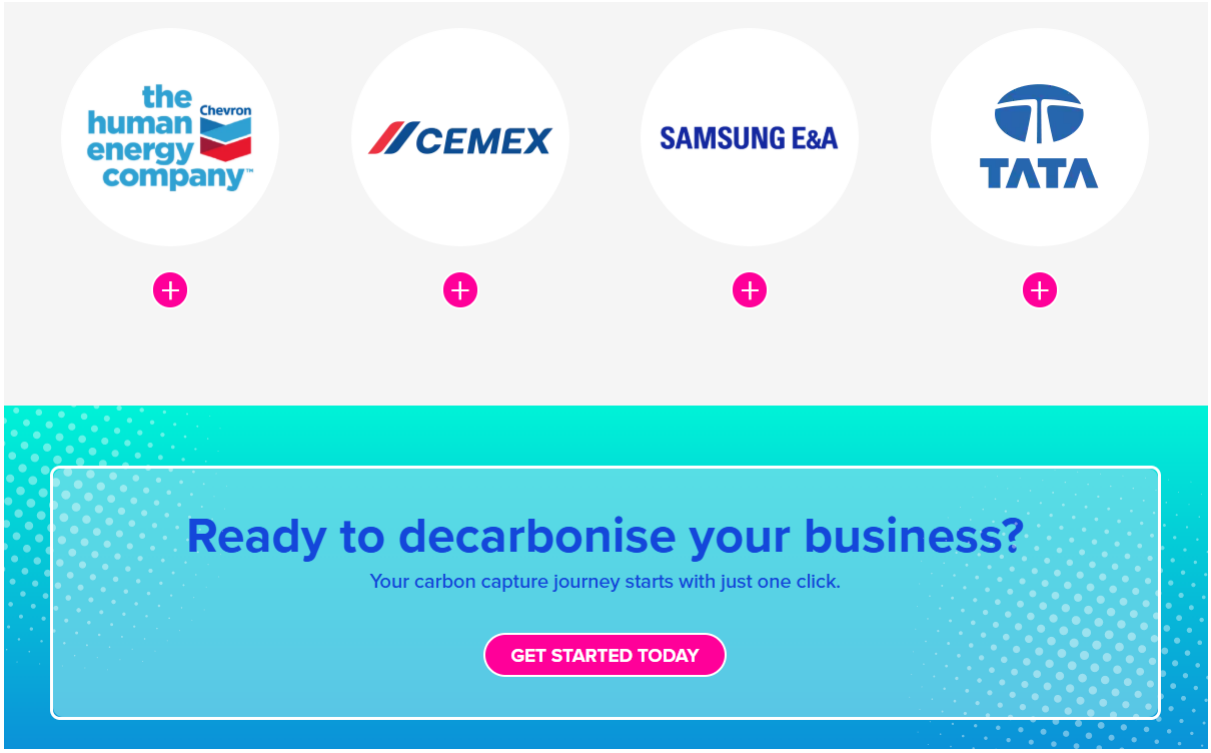
- On the Fervo about page, the partners and investors are displayed with links to each one. They also had a ‘commercial partnership opportunities‘ form on the contact page




- Invenergy had their affiliates displayed on a collapsed list in their home page header and an organisation-based contact page




- Heliogen used a 'partnership' option in their contact form




- Carbon clean had an area displaying their partners and a 'get started today' link beneath it to decarbonise your business. Presumably, a business can become and affiliate through the 'contact an expert' on the home page

## Industry guidelines and regulations:

Green technology focuses on pollution control, energy efficiency and renewable energy. There are many key pieces of legislation which apply to this:

- EU Industrial Emissions Directive (IED)
- UK Streamlined Energy & Carbon Reporting (SECR)
- UK Environmental Bill (upcoming)
- UK Plastic Packaging Tax
- UK Energy Act 2023
- UK ZEV Mandate – ban new petrol/diesel car sales by 2030
- EU Net-Zero Industry Act

For the company's operations to be optimal, they must abide by these regulations. For example, due to the ZEV Mandate, it could be best to use electric vehicles from an early point. There are also many pieces of legislation which apply to a potential website:

- WCAG 2.2
- UK GDPR 2018
- DMCCA 2024
- PECR
- Equality Act 2010
- Copyright, Designs and Patents Act